

ADR-004: Progressive Replay Format and Manifests

Project Genesis Research Repository

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Status

Approved

Context

Large simulation runs can exceed hundreds of megabytes in `\textttreplay.json` format, making single-load delivery to browsers slow and memory-intensive.

Decision

Avoid complete file truncation where possible. We will design the replay system to package replays into chunked files (e.g., `\textttchunk\_00001.bin`, `\textttchunk\_00002.bin`) referenced by a `\textttmanifest.json`.

Consequences

Enables progressive loading of tick data as the user scrubs through the timeline, preventing browser crashes and enabling long-duration simulation replays.